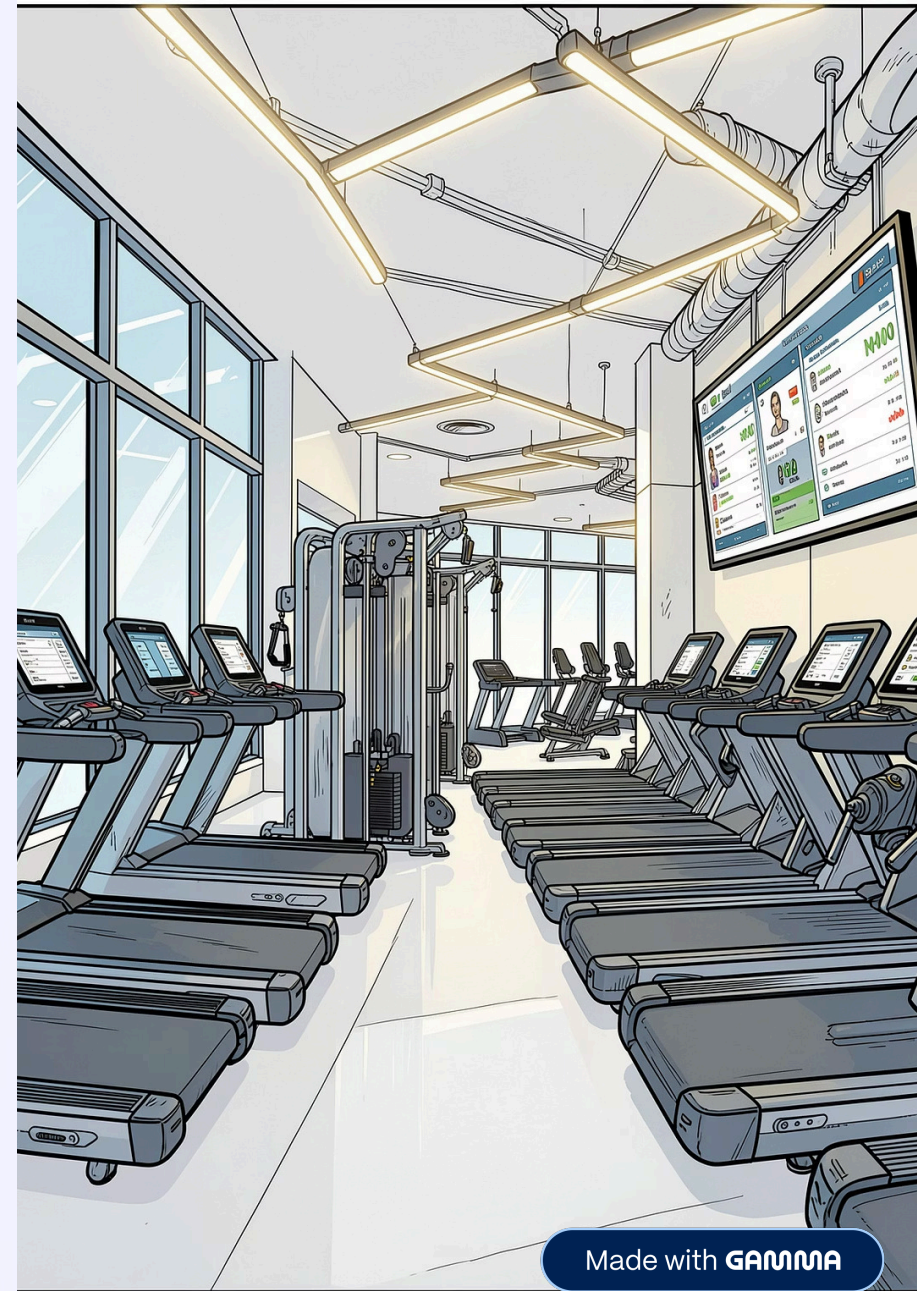


PART 4

Use Cases, Business Rules & Acceptance Criteria

This section of the GymX Software Requirements Specification defines the primary interactions between users and the system, the rules that govern system behavior, and the measurable acceptance criteria that each module must satisfy. It serves as the definitive reference for developers, QA engineers, product managers, and stakeholders involved in the GymX development lifecycle.



Use Case Specifications — Overview

This section describes the eight primary use cases that define how each user role interacts with the GymX system. Use cases are the cornerstone of functional requirements — they translate business needs into concrete, testable user flows. Each use case specifies actors, preconditions, main flows, alternative flows, and postconditions, ensuring unambiguous implementation guidance for the development team.

01

UC-001 User Login

All roles — secure authentication via JWT

02

UC-002 Register New Member

Receptionist — onboarding with profile and plan

03

UC-003 Renew Membership

Receptionist — plan selection and invoicing

04

UC-004 QR Check-In

Member — automated attendance via QR scan

05

UC-005 Create Workout Plan

Coach — exercise programming for members

06

UC-006 Create Diet Plan

Nutritionist — calorie-based meal planning

07

UC-007 Process Payment

Accountant — invoice handling and receipts

08

UC-008 Generate Reports

Admin — PDF and Excel export capabilities

User Login & Member Registration

UC-001 User Login

Actors: All eight user roles – Super Admin, Admin, Branch Manager, Receptionist, Coach, Nutritionist, Accountant, and Member.

Preconditions: User account exists and is in an active state.

Main Flow:

1. User opens the login page
2. User enters email and password
3. System validates credentials against the database
4. JWT token is generated upon successful validation
5. User is redirected to their role-specific dashboard

Alternative Flows: Invalid credentials display an error; disabled accounts are rejected; expired passwords prompt a reset flow.

Postconditions: User session is established and login activity is logged for audit purposes.

UC-002 Register New Member

Actor: Receptionist (must be authenticated)

Main Flow:

1. Open the member registration module
2. Enter personal information (name, contact, DOB)
3. Upload a profile image
4. Select an appropriate membership plan
5. Save the member record
6. System auto-generates a unique Member ID
7. Send a welcome notification to the new member

Postconditions: Member account is created successfully and is immediately available for check-in and plan assignment.



The Member ID generated at step 6 is immutable and serves as the primary reference key across all GymX modules.

Membership Renewal & QR Check-In

UC-003 Renew Membership

Actor: Receptionist

The renewal flow ensures members can extend their gym access seamlessly, with full financial traceability through invoice generation.

1. Search for the member by name, ID, or phone
2. Select the current membership record
3. Choose the renewal plan (same or upgraded)
4. Receive payment via the configured gateway
5. System generates an invoice automatically
6. Membership expiry date is updated in the database

UC-004 QR Check-In

Actor: Member

The QR check-in system automates attendance tracking, reducing receptionist workload and providing real-time gym occupancy data.

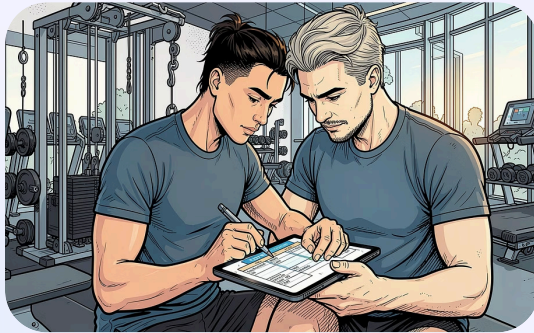
1. Member scans their unique QR code at entry
2. System validates membership status and expiry
3. Attendance record is created with timestamp
4. Success message is displayed to the member

Alternative Flows:

- Expired membership – access denied with renewal prompt
- Duplicate check-in – warning displayed, no duplicate record
- Invalid QR code – error message shown

Workout & Diet Plan Creation

These two use cases represent the core value proposition of GymX's coaching and nutrition modules – enabling personalized, structured plans to be created by qualified staff and delivered directly to members through the platform.



UC-005 Create Workout Plan

Actor: Coach

1. Select the target member from the roster
2. Create individual exercises with descriptions
3. Set repetitions, sets, and rest intervals
4. Define plan duration and schedule
5. Save the workout plan to the system
6. Notify the member via in-app notification



UC-006 Create Diet Plan

Actor: Nutritionist

1. Select the target member from the roster
2. Calculate daily caloric requirements based on member profile
3. Create structured meals (breakfast, lunch, dinner, snacks)
4. Save the diet plan to the system
5. Notify the member of the new plan availability



Both plans support incremental updates – coaches and nutritionists can modify existing plans without creating new records, preserving historical data for progress tracking.

Payment Processing & Report Generation

UC-007 Process Payment

Actor: Accountant

Every financial transaction in GymX must be fully auditable. This use case ensures that payments are recorded accurately, invoices are generated automatically, and receipts are available for download.

1. Select the outstanding invoice
2. Choose the payment method (cash, card, transfer)
3. Confirm payment through the gateway
4. System generates a downloadable receipt
5. Financial records are updated in real time

UC-008 Generate Reports

Actor: Admin

Reporting is a critical administrative function, providing stakeholders with actionable insights across all modules. Reports can be filtered by date range and exported in industry-standard formats.

1. Select the report type (attendance, revenue, membership)
2. Define the date range for the report
3. Generate the report with live data
4. Export to PDF for distribution
5. Export to Excel for further analysis

Report types include: membership summaries, attendance logs, financial statements, and inventory status – all accessible from a single admin interface.

Business Rules

The GymX system enforces a strict set of business rules across five domains. These rules are non-negotiable constraints that must be validated at both the application and database levels to ensure data integrity, regulatory compliance, and consistent user experience.

Membership Rules

- Every member must have exactly one active membership at any time
- Expired memberships cannot access gym facilities
- Frozen memberships extend expiration dates proportionally
- Memberships can only be renewed before deletion – not after

Attendance Rules

- One active check-in per member at a time – no concurrent sessions
- Check-out requires a preceding valid check-in record
- Attendance history is immutable – records cannot be deleted

Payment Rules

- Every payment transaction generates a corresponding invoice
- Payment amounts cannot be negative under any circumstance
- Refunded payments remain in audit history for traceability

User Rules

- Email addresses must be unique across all user roles
- Phone numbers must be unique to prevent duplicate accounts
- All passwords must be encrypted using industry-standard hashing
- Deleted users become inactive – permanent deletion is prohibited

Inventory Rules

- Stock levels cannot become negative – system must prevent overselling
- Equipment maintenance history cannot be removed – full audit trail required
- Supplier information must remain available even after contract end

Acceptance Criteria

Each GymX module must satisfy measurable, verifiable acceptance conditions before it can be considered complete. These criteria form the basis of QA test plans and are the definitive gate for sprint sign-off. Every criterion must be demonstrable through manual or automated testing.



Authentication

- User logs in successfully with valid credentials
- Invalid password displays a clear error message
- Expired token prompts a refresh or re-login



Members

- Member created successfully with all required fields
- Duplicate email is rejected at validation
- Membership is assigned at registration
- Profile updates persist correctly



Attendance

- QR Check-In functions correctly on valid membership
- Duplicate attendance is prevented automatically
- Attendance history is stored and retrievable



Workout

- Coach creates and saves workout plans
- Member can view assigned workout plans
- Progress updates are recorded correctly



Nutrition

- Diet plans are assigned to members by nutritionists
- Calories are calculated based on member profile
- Meal schedules are displayed correctly



Payments

- Invoice is generated for every payment
- Payment is recorded in financial ledger
- Receipt is downloadable by the member



Reports

- Reports generate accurately for selected date ranges
- Export to PDF functions without data loss
- Export to Excel preserves all columns and formatting



Inventory

- Products are added with all required attributes
- Stock levels update on every transaction
- Low stock alerts are generated at threshold

System Constraints & Assumptions

Technology Stack Constraints

The GymX project is bound to the following technology choices, which define the architectural foundation and constrain all implementation decisions:

→ Backend

Django + Django REST Framework – Python-based, batteries-included framework for rapid, secure API development

→ Database

PostgreSQL – enterprise-grade relational database with full ACID compliance and advanced querying capabilities

→ Frontend

React.js – component-based JavaScript framework for responsive, single-page application experience

→ Infrastructure

Docker – containerized deployment ensuring environment parity across development, staging, and production

→ Security

JWT Authentication – stateless, token-based authentication with refresh token rotation for session security

The application must support modern browsers only (Chrome, Firefox, Safari, Edge – latest two versions). Internet connectivity is required for all core operations.

Project Assumptions

The following assumptions underpin the GymX requirements. If any assumption proves invalid during development or deployment, the requirements must be revisited and updated accordingly.



Internet Access

All users – staff and members – have reliable internet connectivity to access the system



Staff Training

All GymX staff users receive adequate training before system go-live



Email Services

An SMTP or third-party email service is configured and operational for notifications



QR Scanners

Hardware QR scanners or mobile devices with camera access are available at all entry points



Payment Gateway

A payment gateway integration (e.g., Stripe, PayPal) is available and configured



Database Uptime

The PostgreSQL server maintains high availability with minimal downtime SLA

Traceability Matrix & Conclusion

Requirements Traceability Matrix

Each functional requirement is mapped to its corresponding module and priority level, ensuring full coverage from specification through to testing and delivery. This matrix enables impact analysis when requirements change and supports audit compliance.

Requirement ID	Module	Priority
FR-001	Authentication	High
FR-002	User Management	High
FR-003	Member Management	High
FR-004	Membership	High
FR-005	Attendance	High
FR-006	Workout	Medium
FR-007	Nutrition	Medium
FR-008	Classes	Medium
FR-009	Payments	High
FR-010	Inventory	Medium
FR-011	Equipment	Medium
FR-012	Notifications	Low
FR-013	Dashboard	High
FR-014	Reports	High
FR-015	AI Assistant	Future

Conclusion

This Software Requirements Specification – Part 4 – defines the complete functional landscape of the GymX Gym Management System. Together with preceding parts, it provides developers, testers, designers, and stakeholders with a comprehensive, unambiguous reference for building a secure, scalable, and enterprise-grade platform.

 Secure JWT authentication, encrypted passwords, and immutable audit trails	 Scalable Docker containerization and PostgreSQL support enterprise-level growth
 Verifiable Measurable acceptance criteria enable rigorous QA and sprint sign-off	 Traceable Full requirement-to-module traceability supports change management